

ELEC ENG 2411 Ex: 2

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Submitted to:
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Part 4 Step 1

We selected our base components for our RC circuit and found our components were within tolerance. The internal resistance was combined with our resistor for a total resistance of 1041.3Ω .

$R (\Omega)$	$C (F)$	$R_{in} (\Omega)$	$R' = R_{in} + R (\Omega)$
991.3	0.000000112	50	1041.3

Table 1: Values for component pieces and total circuit resistance.

Part 4 Step 3

With a 100Hz sinusoid, we find that the effect of the system on input signal is negligible with no noticeable difference on phase shift and V_{pp} as seen on channel 2 in figure 1. As the frequency of channel 1 increases, we see that the phase shift and amplitude of channel 2 change drastically. A 1kHz sinusoid results in a phase shift of -29.52° and an amplitude of $1.84V$ shown in figure 2. A 10kHz sinusoid results in a phase shift of -74.02° and an amplitude of $336mV$ shown in figure 3. Based on these effects, we can consider this circuit to be a low pass filter. We can determine this by looking at figure 3 where the output signal appears to be increasing as the input signal decreases. This indicates that the circuit is passing “low” input.

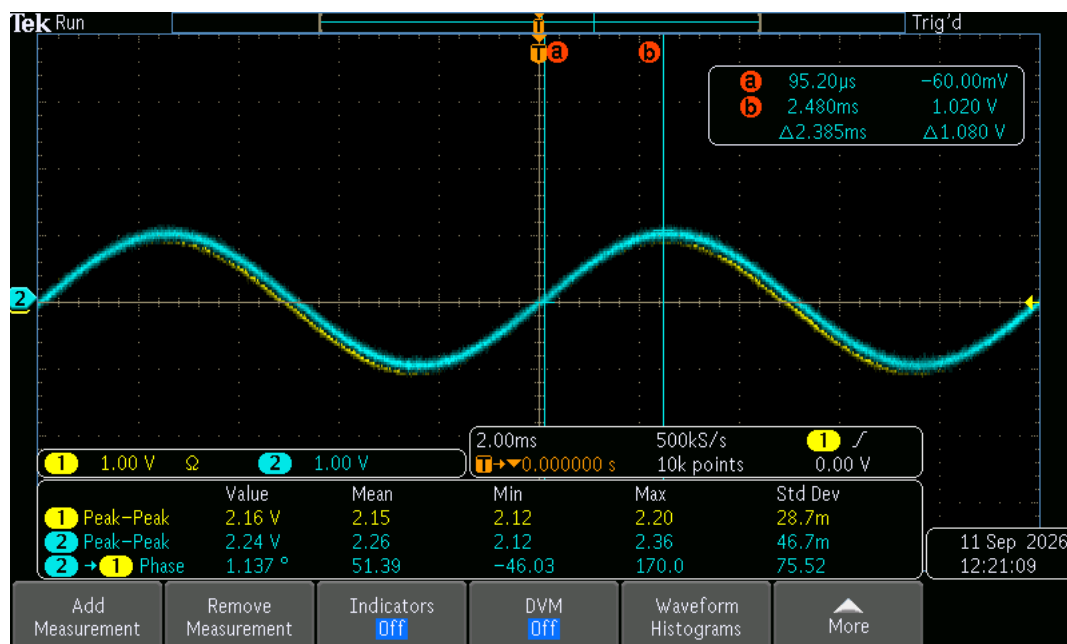


Figure 1: 100Hz Sinusoid

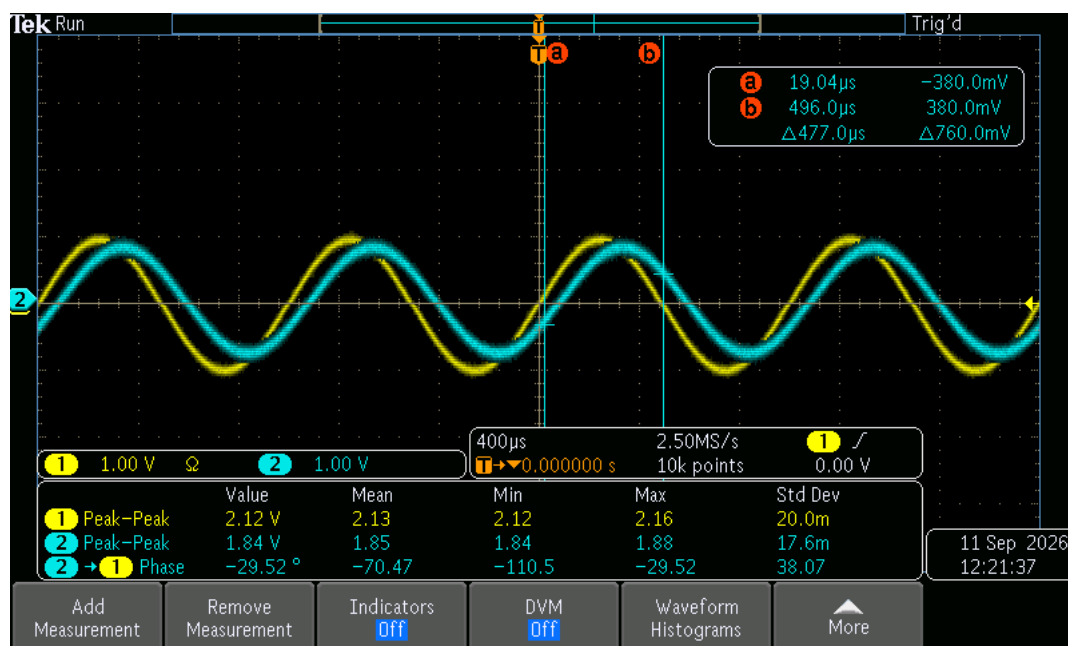


Figure 2: 1kHz Sinusoid

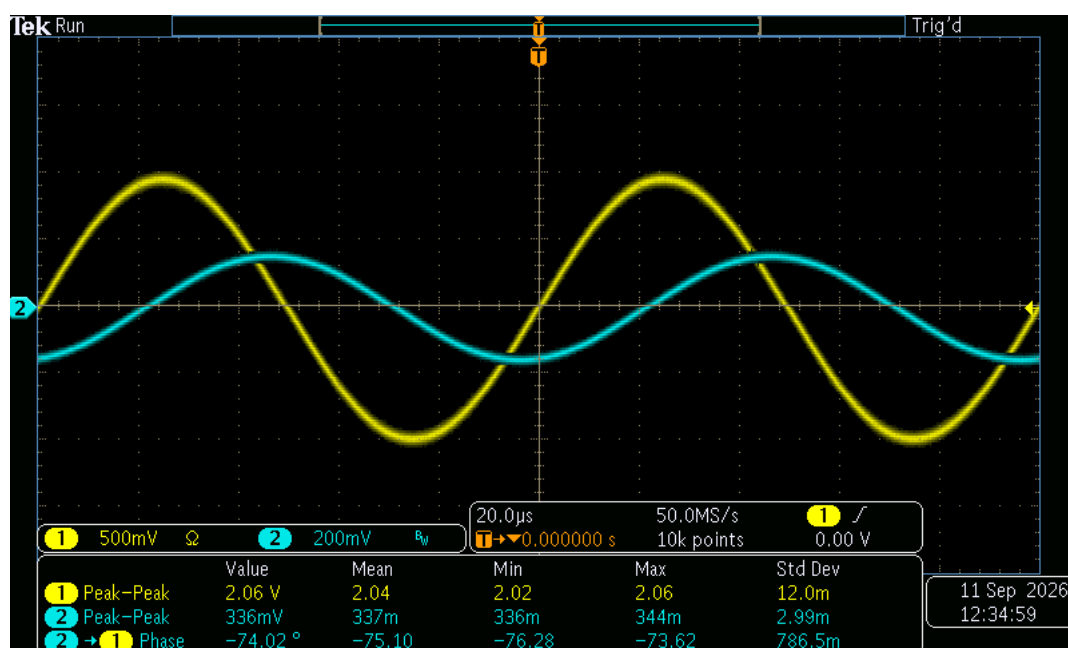


Figure 3: 10kHz Sinusoid

Frequency (kHz)	Channel 1 V_{pp}	Channel 2 V_{pp}	$\theta_{2,1}^\circ$
0.100	2.16	2.24	1.137
1.0	2.12	1.84	-29.52
10.0	2.06	0.336	-74.02

Table 2: Increasing frequency values with associated channel V_{pp} and phase shift.

Part 4 Step 4

To find the time constant we consider our system where $x(t)$ is the input and $y(t)$ is the output.

$$\frac{dy(t)}{dt} + \frac{1}{RC}y(t) = \frac{1}{RC}x(t)$$

We assume the form $y(t) = Ae^{\lambda_1 t} + Be^{\lambda_2 t}$. We then write out the characteristic equation and solve for λ . Then we substitute λ into our solution.

$$\begin{aligned}\lambda + \frac{1}{RC} &= 0 \\ \lambda_1 &= -\frac{1}{RC} \\ y(t) &= Ae^{-\frac{t}{RC}}\end{aligned}$$

From our solution, we find the time constant $\tau = RC$. Using experimental values for R and C we find the experimental time constant should be 1.17×10^{-4} . The ratio of the rise time over the time constant is 1.12.

Rise Time (μs), t_r	Time constant (μs), τ	$\frac{t_r}{\tau}$
131.3	117	1.12

Table 3: Experimental rise time, time constant, and ratio of rise time and time constant.

Part 4 Step 5

Both the 100Hz signal and the 1kHz signal show no distortion of the output signal and we see on both figure 4 and figure 5 have the same signals within margin of error. The 10kHz signal shown in figure 6 shows drastic distortion. We see that V_{pp} of channel 2 is 600mV with the start of the signal showing above the low signal of channel 1 indicating that the capacitor does not have enough time to discharge before a high signal enters the system so there is still residual potential in the capacitor. The capacitor also does not have enough time to charge before the high signal leaves and does not reach the full amplitude of the input. Given that our time constant is the time it takes to reach 63% of the full potential, a 10kHz square has a period of $100\mu s$ and a high or low signal has a time of $50\mu s$, we can determine that our signal is too “quick” since $50\mu s < 117\mu s$.

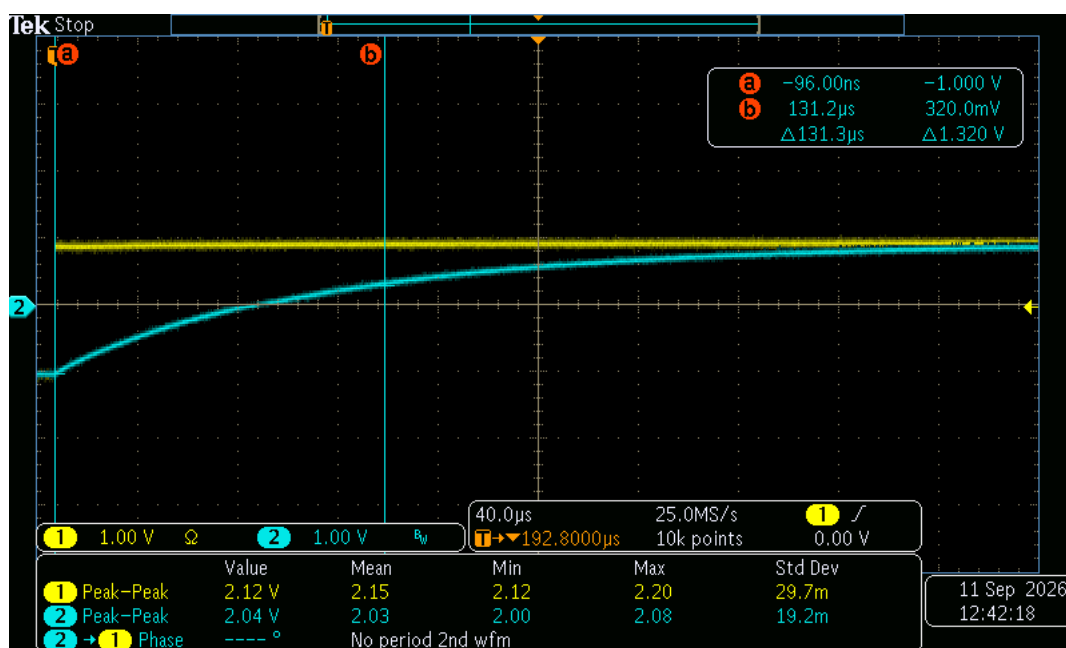


Figure 4: 100Hz Square

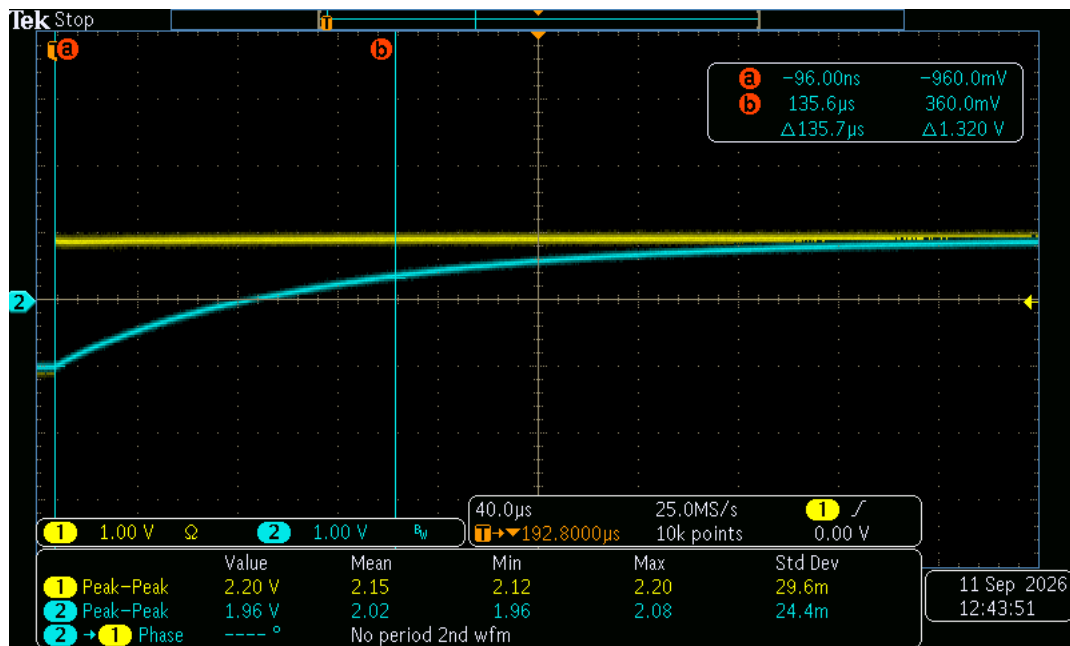


Figure 5: 1kHz Square

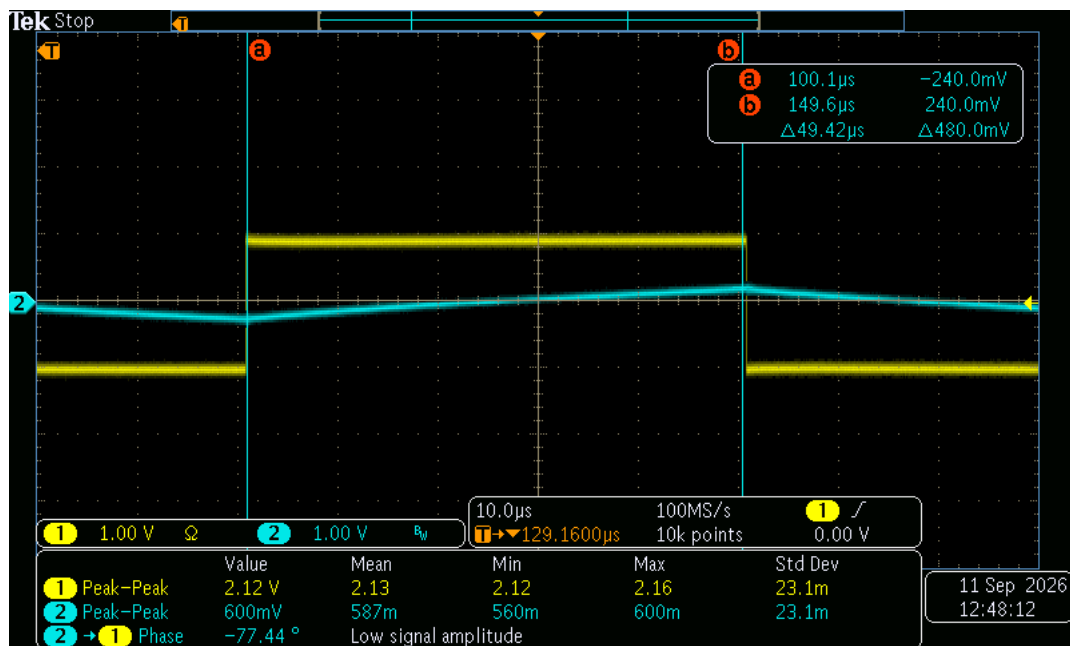


Figure 6: 10kHz Square

Frequency (kHz)	Channel 1 V_{pp}	Channel 2 V_{pp}
0.100	2.12	2.04
1.0	2.20	1.96
10.0	2.12	0.600

Table 4: Frequency of input signal and voltage peaks for input and output signals.

Part 5 Step 6

We use MATLAB Simulink software to model our system on our computer. We ran our simulation with a runtime of $3/freq$ so each runs for three periods. To find the phase shift, we take the time change between signal peaks and multiply it by 2π over the period. We found that our simulation appears to behave as a low pass filter with behavior that increases as the frequency increases. The changes in amplitude and phase is comparable to that of our experimental results.

$$\Delta\theta = \Delta t \frac{2\pi}{T}$$

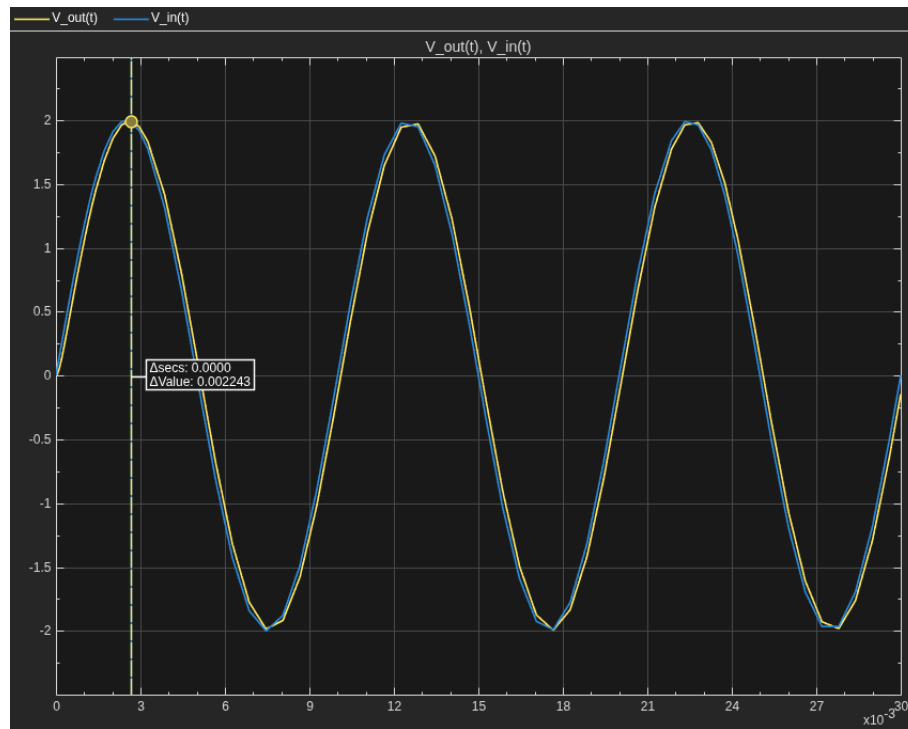


Figure 7: 100Hz Sinusoid

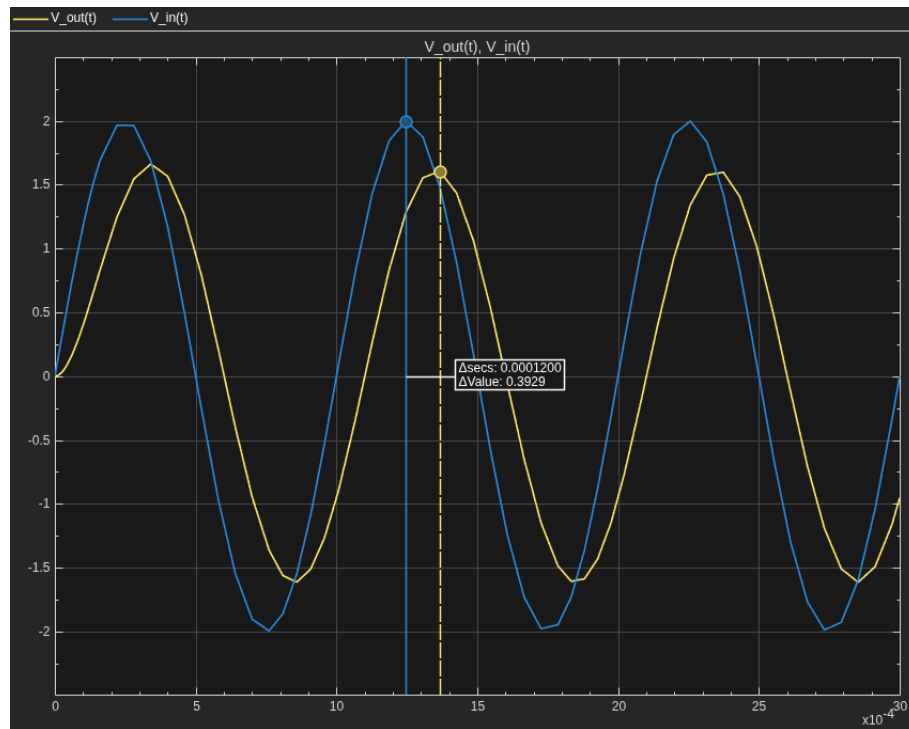


Figure 8: 1kHz Sinusoid

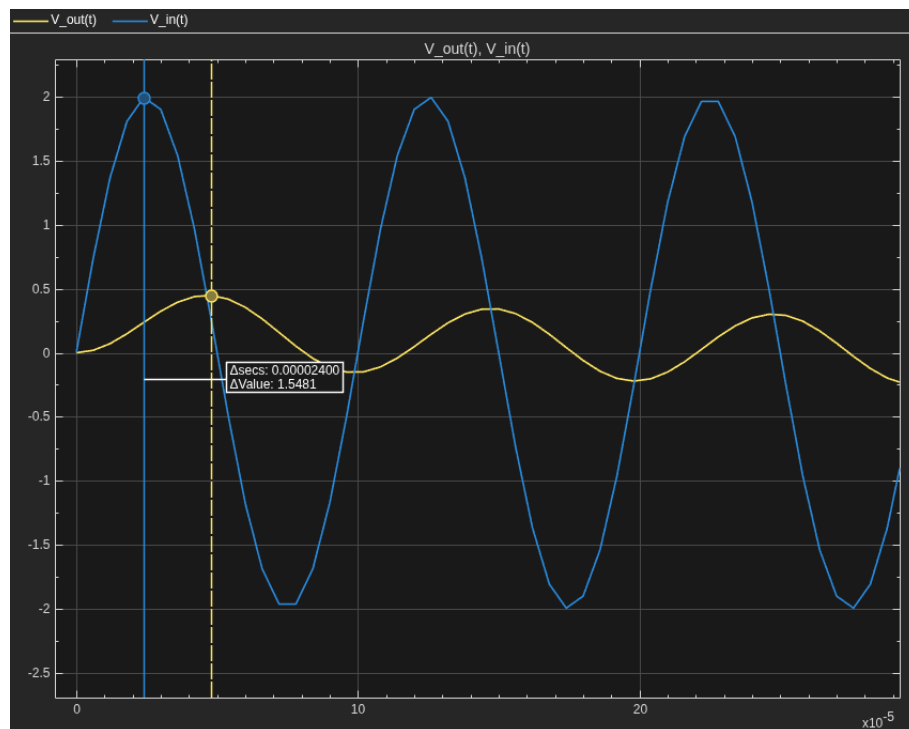


Figure 9: 10kHz Sinusoid

Frequency (kHz)	$V_{in,pp}$	$V_{o,pp}$	$\Delta\theta(rad)$
0.100	1.996	1.994	0
1.0	1.996	1.661	0.75
10.0	1.996	0.449	1.51

Table 5: Theoretically modeled frequencies with their associated V_{pp} and phase shift.

Part 5 Step 7

Using the standard percent error formula, we found the percent error of our rise time to be 1.29%. We also found the percent error of our ratio to be 1.79%.

Rise Time (μs), t_r	Time constant (μs), τ	$\frac{t_r}{\tau}$
133.0	117	1.14

Table 6: Theoretically modeled rise time, time constant, and ratio of rise time over time constant.

Part 5 Step 8

We use MATLAB Simulink software to model our system on our computer. We ran our simulation with a runtime of $3/freq$ so each runs for three periods. We found that the Simulink output matches that of our physical experiment though the behavior is more pronounced in the simulation.

For our sinusoidal model, we found that at the 100Hz signal the peak to peak voltages and phase shift are roughly the same. For the 1kHz signal, the peak to peak voltages in the experimental case are roughly 10% higher than the model. The phase shift in our experiment is half what is shown in the model. For the 10kHz signal, the peak to peak voltages in the experimental case are roughly 25% higher than the model. The phase shift is also roughly 14% higher in the experimental case than the model.

For our square model, we found that at the 100Hz signal, the peak to peak voltages were roughly the same. For the 1kHz signal, the peak to peak voltages were also roughly the same. For the 10kHz signal, the voltage peaks for the output were roughly 30% higher in the experiment compared to the model.

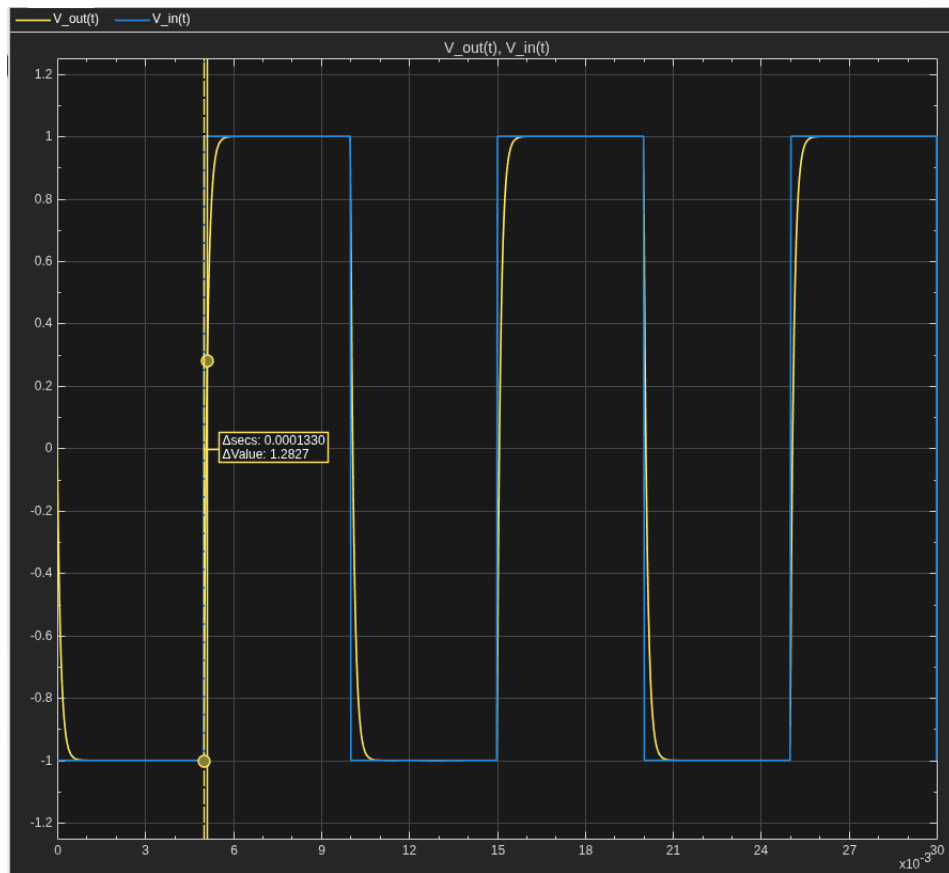


Figure 10: 100Hz Square

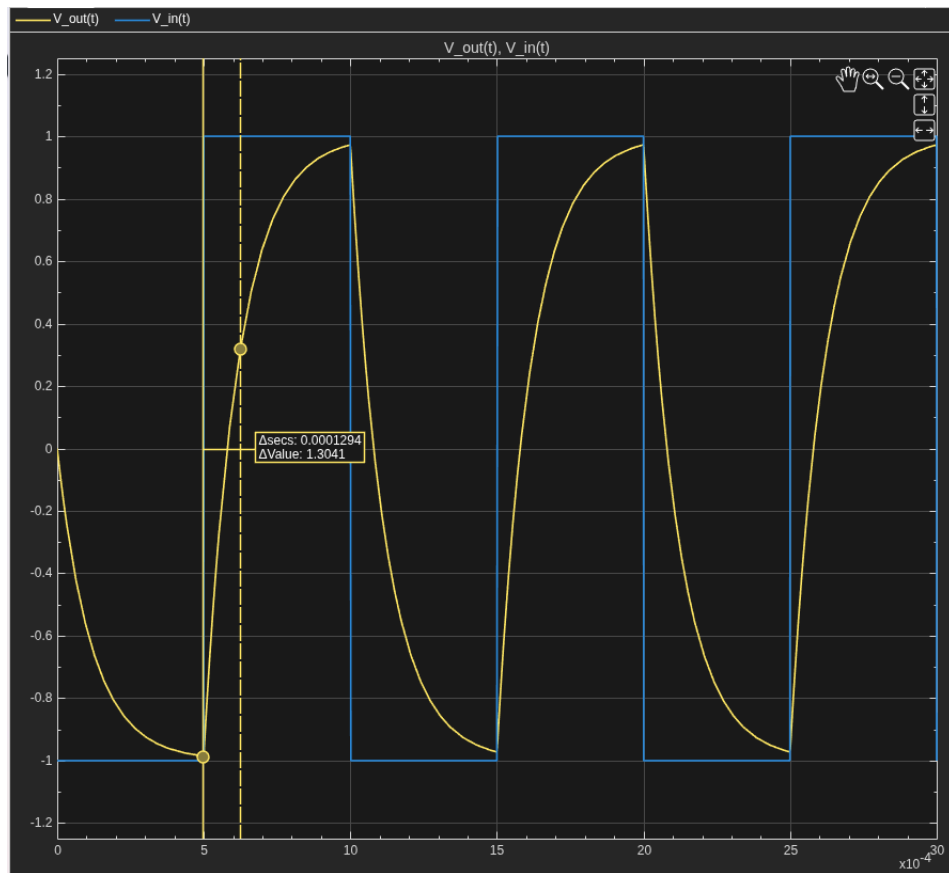


Figure 11: 1kHz Square

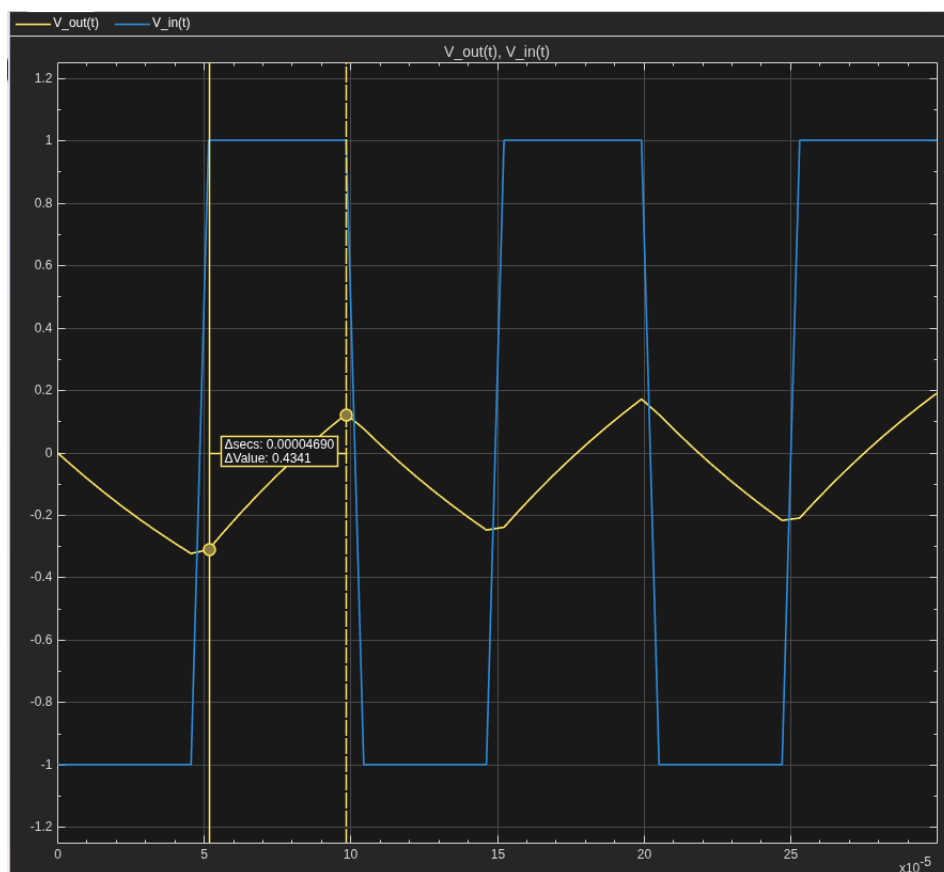


Figure 12: 10kHz Square

Frequency (kHz)	Channel 1 V_{pp}	Channel 2 V_{pp}
0.100	2.0	2.0
1.0	2.0	1.94
10.0	2.0	0.43

Table 7: Theoretically modeled frequency of input signal and voltage peaks for input and output signals.